

Bruno Mingoti

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EDUCATION

RWTH Aachen University

B.Sc. Mechanical Engineering - Exchange Student (GPA n/a)

Aachen, Germany

08.2024 – 12.2025

- Student Assistant / *Hilfswissenschaftler* at *Werkzeugmaschinenlabor WZL*

Federal University of Santa Catarina (UFSC)

B.Sc. Electrical Engineering (GPA 8.91/10.0)

Florianopolis, Brazil

08.2020 – 07.2027 (expected)

- Relevant courses include machine learning, statistics, advanced calculus, image processing and linear algebra.
- Projects: lithium battery parameter identification (Python); acute lymphoblastic leukemia cell classification (93.88% weighted F1-score with InceptionV3 ensemble); H&E tissue cell segmentation and estimation (OpenCV, scikit-image).

WORK EXPERIENCE

Werkzeugmaschinenlabor WZL der RWTH Aachen

Student Assistant (Hilfswissenschaftler – HiWi)

Aachen, Germany (On-site)

08.2024 – 12.2025

Leading research institute on machine tools and production engineering.

- Completed bachelor's thesis on "Filtered backprojection combined with deep learning-based sinogram denoising for industrial CT reconstruction"; implemented two GPU-accelerated reconstruction pipelines and benchmarked against institute software, improving quality and performance.
- Built backend and frontend services for automated data-stream management (device/producer/consumer matching). ([Automated Data Stream Management Software-Stack](#)).
- Implemented database operations for storing and signing measurement datasets for Digital Calibration Certificates (DCC). ([DCC Generator and Validator Webpage](#)).
- Supported industrial metrology projects: optical measurements, analysis, and reporting.
- Assistance in hosting the 3DMC - Europe's finest metrology event.

CERTI Foundation

Co-op student at NEO Empresarial

Florianopolis, Brazil (On-site)

03.2022 – 07.2024

Innovative program designed to offer distinguished education to engineering students, through technical and managerial projects in partnership with internationally recognized companies, such as WEG S.A., Embraco/Nidec Corporation and Landis+Gyr AG.

- Collaborated with engineering teams and industry partners to gather requirements, present technical findings, validate prototypes, and translate business challenges into data-driven solutions.
- Researched AI methods for few-shot learning in collaboration with CERTI Foundation.
- Developed STM32 firmware (C, STM32CubeIDE) for molded-case circuit breaker calculations (tripping curve) and communication via I2C/SPI for WEG S.A.
- Developed laptop failure detection using YOLO (defects such as stains and dents) for Lenovo.
- Built an Electron + Modbus supervisory application for stress relief in WEG's Boiler Area (estimated €40k/year savings).
- Automated vibration-spectrum analysis and reporting for WEG Motion Fleet Management, transforming sensor data into standardized insights for engineering decision-making and deploying the solution into production.
- Development of object detection and object tracking models to identify objects on the sea through aerial images, with a focus on identifying wrecks, vessels and lost people. The project is a partnership with CERTI Foundation in a cooperation between Brazil and Sweden in Aeronautics, which seeks to use AI to transform search and rescue missions around the world.
- Instructor in the group's data science training program for new members.

Invest Jr.

Member

Florianopolis, Brazil (Remote)

09.2020 – 06.2021

Nonprofit organization managed by students.

- Designed and delivered technical training and digital solutions focused on financial analysis and business management. Produced instructional content on Excel, Power Query, and VBA; led workshops and courses on investments and personal finance; and developed automated spreadsheet systems, dashboards, and reporting tools to optimize financial tracking, sales management, and decision-making for clients.

PUBLICATIONS

- **Small Object Detection in Drone-Captured Images for Maritime Surveillance**, SIBGRAPI 2025
- **Filtered backprojection combined with deep learning-based sinogram denoising for improved industrial CT reconstruction**, UFSC 2025
- **Experimental Identification of a Simplified Lithium-Ion Battery Model (original in Portuguese)**, CBMAG 2024

LANGUAGES AND SKILLS

LANGUAGES: Portuguese – Native; English – C1.

HARD SKILLS: Programming (Python, TypeScript, React, Node.js); APIs (FastAPI); deployment (Docker); version control (Git); Data/ML (pandas, NumPy, SciPy, Matplotlib, Plotly, PyTorch, TensorFlow, OpenCV, scikit-image); Excel, SQL

SOFT SKILLS: logical thinking, interpersonal skills, self-taught, resilience, teamwork, stakeholder communication